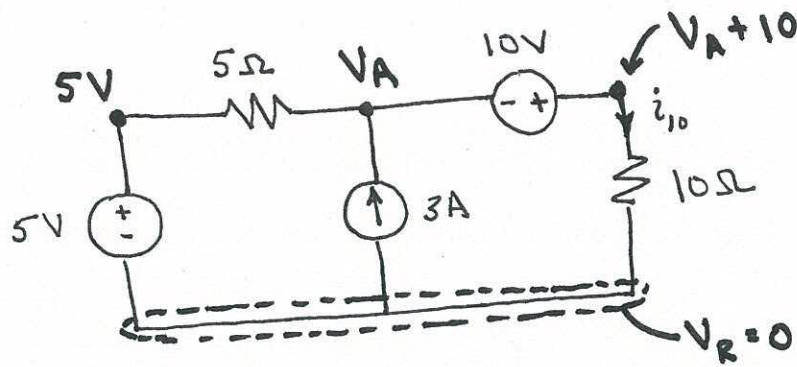


3.1



One independent voltage - V_A

KCL at "A":

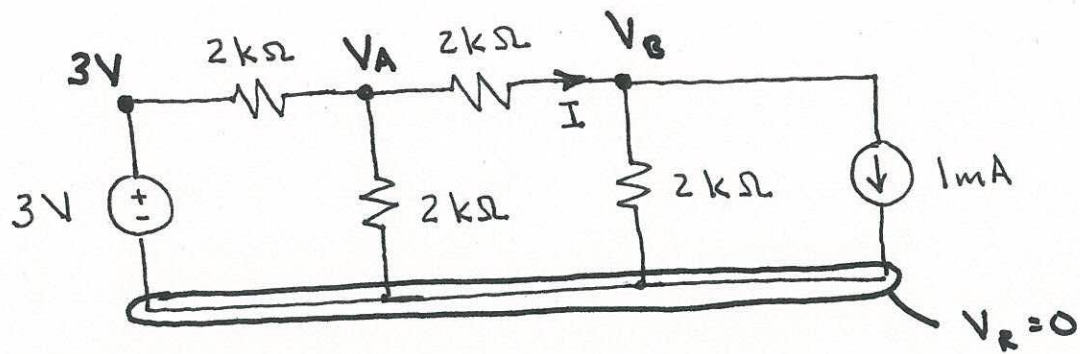
$$\frac{V_A - 5V}{5\Omega} - 3A + \frac{V_A + 10}{10\Omega} = 0$$

$$2V_A - 10 - 30 + V_A + 10 = 0 \Rightarrow 3V_A = 30V \Rightarrow V_A = 10V$$

Ohm's Law: $i_{10} = \frac{V_A + 10}{10\Omega} = \frac{20}{10} \Rightarrow \boxed{i_{10} = 2A}$

In the direction shown.

3.2



Two independent voltages - $V_A \neq V_B$

Node A: $\frac{V_A - 3V}{2k\Omega} + \frac{V_A}{2k\Omega} + \frac{V_A - V_B}{2k\Omega} = 0 \Rightarrow 3V_A - V_B = 3$

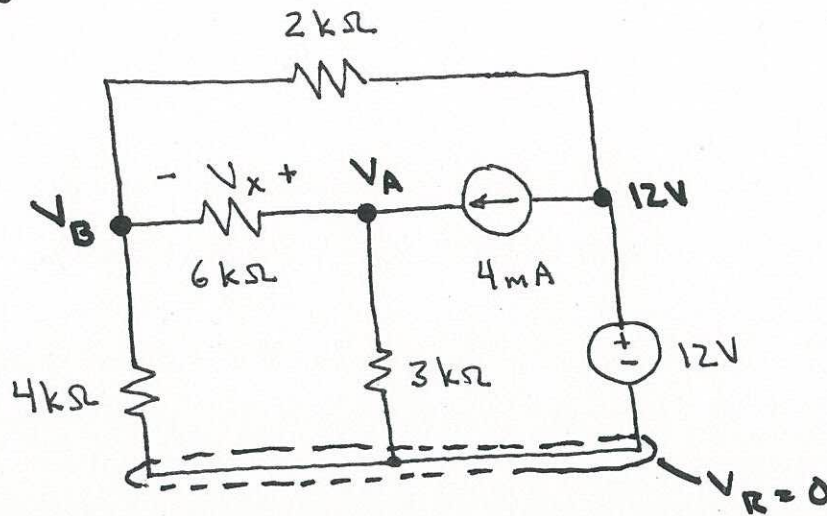
Node B: $\frac{V_B - V_A}{2k\Omega} + \frac{V_B}{2k\Omega} + 1mA = 0 \Rightarrow 2V_B - V_A = -2$

$$V_A = 0.8V$$

$$V_B = -0.6V$$

$$I = \frac{V_A - V_B}{2k\Omega} = \frac{0.8 - (-0.6)}{2k\Omega} \Rightarrow I = 0.7mA$$

3.3



Two independent
voltages, $V_A \neq V_B$

Node A: $\frac{V_A - V_B}{6k\Omega} + \frac{V_A}{3k\Omega} - 4mA = 0 \Rightarrow 3V_A - V_B = 24$

Node B: $\frac{V_B}{4k\Omega} + \frac{V_B - V_A}{6k\Omega} + \frac{V_B - 12}{2k\Omega} = 0 \Rightarrow 11V_B - 2V_A = 72$

$$V_A = 10.84V$$

$$V_B = 8.52V$$

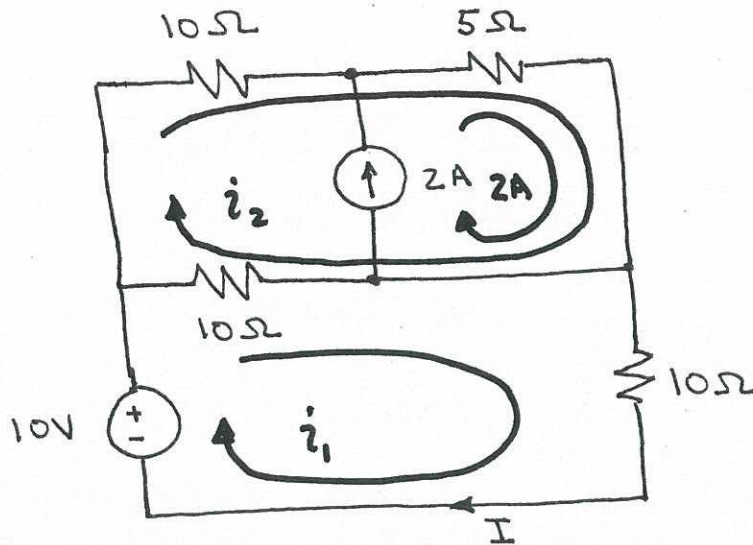
solving

$$V_X = V_A - V_B$$

$$= 10.84 - 8.52$$

$$V_X = 2.32V$$

3.4



2 mesh loops,
1 constrained
loop, as
shown

$$\text{Loop 1: } -10 + 10(i_1 - i_2) + 10i_1 = 0 \Rightarrow i_2 = 2i_1 - 1$$

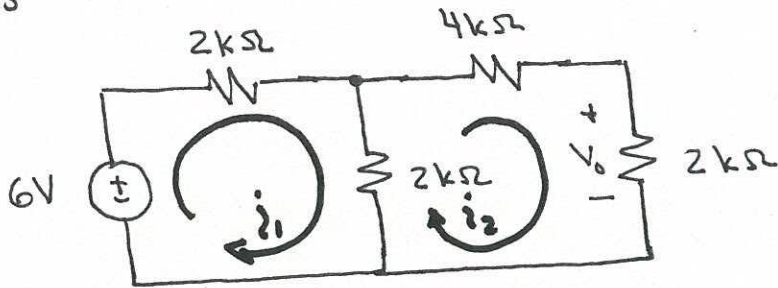
$$\text{Loop 2: } 10i_2 + 5(i_2 + 2) + 10(i_2 - i_1) = 0 \Rightarrow 25i_2 - 10i_1 = -10$$

solving for i_i

$$i_1 = \frac{3}{8} \text{ A}$$

$$I = i_1 \Rightarrow \boxed{I = \frac{3}{8} \text{ A}}$$

3.5



2 mesh currents;
no constraint loops

$$\text{Loop 1: } -6 + 2\text{k}\Omega (i_1) + 2\text{k}\Omega (i_1 - i_2) = 0$$

$$4000 i_1 - 2000 i_2 = 6$$

$$\text{Loop 2: } 2\text{k}\Omega (i_2 - i_1) + 4\text{k}\Omega (i_2) + 2\text{k}\Omega (i_2) = 0$$

$$8000 i_2 - 2000 i_1 = 0$$

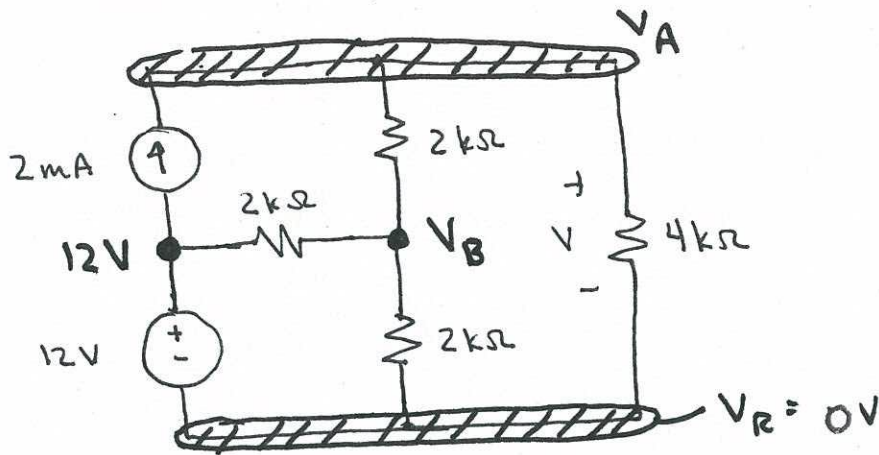
$$i_1 = 1.714 \text{ mA}$$

$$i_2 = 0.429 \text{ mA}$$

SOLVING

$$V_o = 2000 (i_2) \Rightarrow \boxed{V_o = 0.857 \text{ V}}$$

3.6

Nodal Analysis:

2 independent
voltages, $V_A \neq V_B$

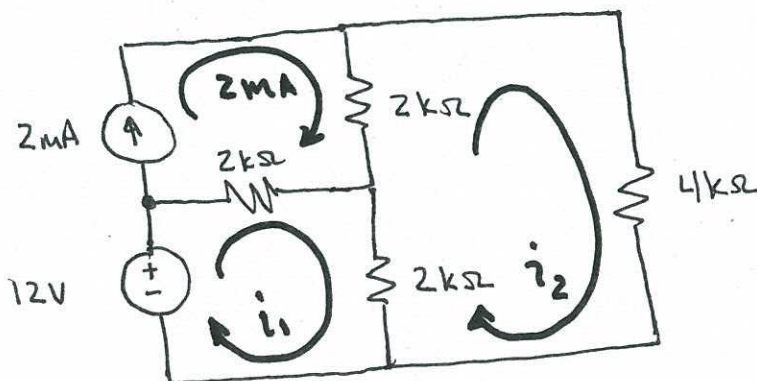
$$\text{Node A: } -2\text{mA} + \frac{V_A - V_B}{2\text{k}\Omega} + \frac{V_A}{4\text{k}\Omega} = 0 \Rightarrow 3V_A - 2V_B = 8$$

$$\text{Node B: } \frac{V_B - 12}{2\text{k}\Omega} + \frac{V_B}{2\text{k}\Omega} + \frac{V_B - V_A}{2\text{k}\Omega} = 0 \Rightarrow 3V_B - V_A = 12$$

$$V_A = 6.857\text{ V}$$

$$V_B = 6.286\text{ V}$$

$$V = V_A = 6.857\text{ V}$$

Mesh AnalysisLoop 1:

$$-12 + (i_1 - 2)2 + 2(i_1 - i_2) = 0$$

$$8 = 2i_1 - i_2$$

Loop 2:

$$2(i_2 - i_1) + 2(i_2 - 2) + 4i_2 = 0$$

$$4i_2 - i_1 = 2$$

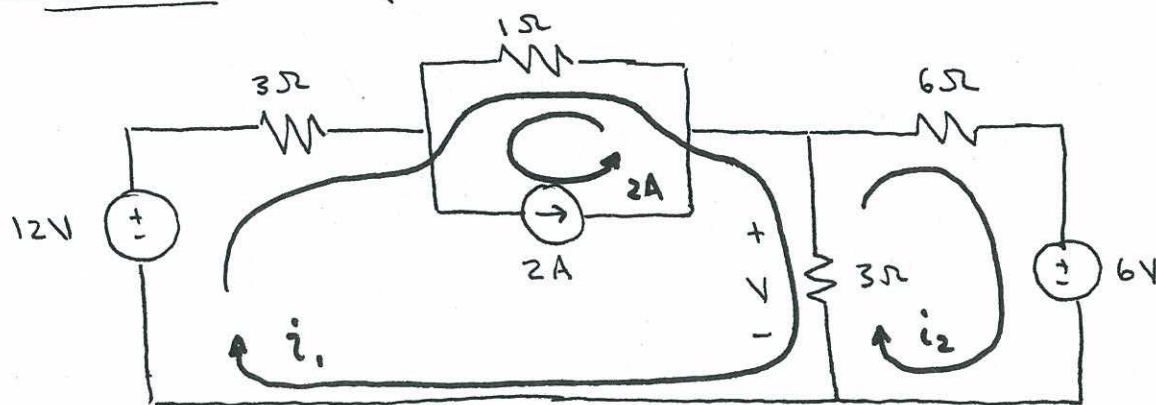
$$i_1 = 4.857\text{ mA}$$

$$i_2 = 1.714\text{ mA}$$

$$V = (4\text{k}\Omega) i_2 \Rightarrow V = 6.857\text{ V}$$

3.7

Two mesh currents, i_1 & i_2 , as shown. One 2A constrained loop:



$$\text{Loop } i_1: -12 + 3\Omega(i_1) + 1\Omega(i_1 - 2A) + 3\Omega(i_1 - i_2) = 0$$

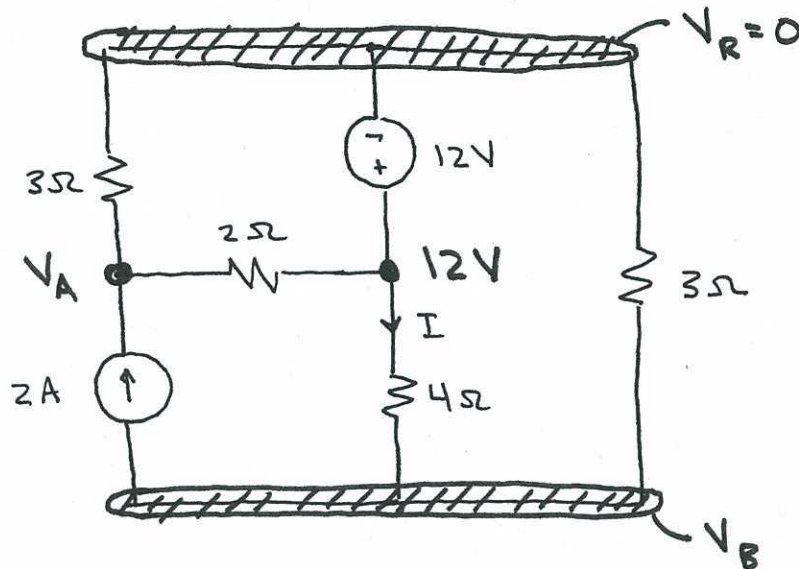
$$\text{Loop } i_2: 3\Omega(i_2 - i_1) + 6\Omega(i_2) + 6V = 0$$

$$V = 3\Omega(i_1 - i_2)$$

Important note: The above solution is not unique! Other valid solutions exist, depending (for one thing) on how you define your constrained loop. You can also define three mesh currents, and write constraint equations explicitly...

3.8

Define reference voltage, two node voltages, and one dependent voltage as shown:



Node A: $\frac{V_A - 0V}{3\Omega} + \frac{V_A - 12V}{2\Omega} - 2A = 0$

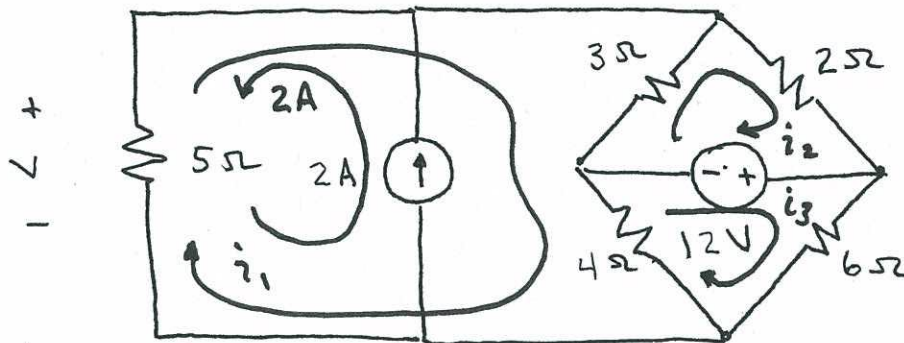
Node B: $2A + \frac{V_B - 12V}{4\Omega} + \frac{V_B - 0V}{3\Omega} = 0$

$$I = \frac{12V - V_B}{4\Omega}$$

The above equations, of course, are not the only possible solution. Other valid solutions exist, depending on (for example) your definitions of reference node and independent nodes.

3.9

Define three mesh currents and one constrained current, as shown below.



$$\text{Loop 1: } (i_1 - 2A)5\Omega + 3\Omega(i_1 - i_2) + 4\Omega(i_1 - i_3) = 0$$

$$\text{Loop 2: } 3\Omega(i_2 - i_1) + 2\Omega(i_2) + 12V = 0$$

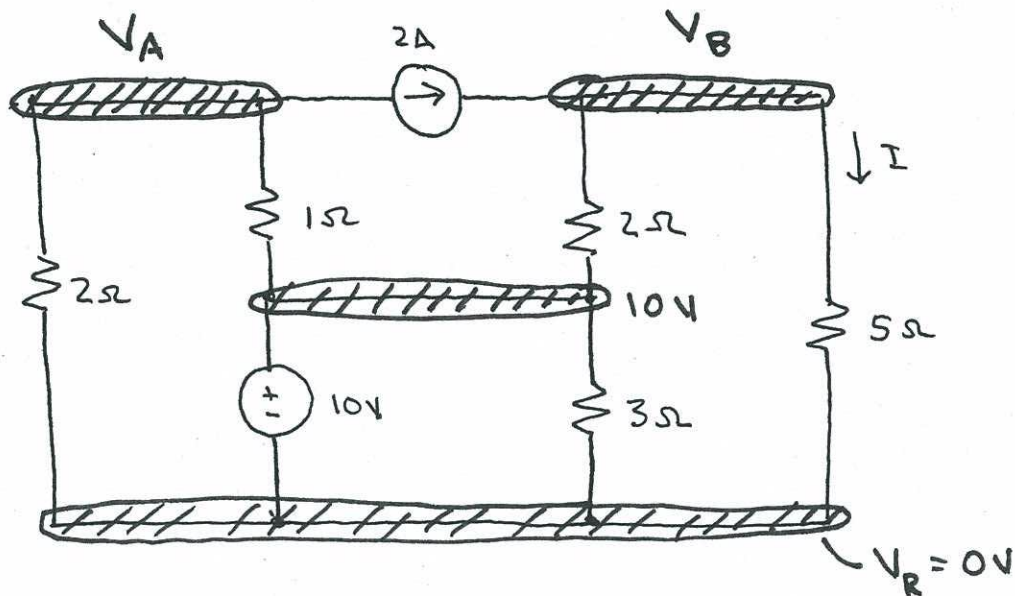
$$\text{Loop 3: } 4\Omega(i_3 - i_1) - 12V + 6\Omega(i_3) = 0$$

$$V = 5\Omega [2A - i_1]$$

Note: Other valid solutions exist, depending upon your definition of mesh loops and constrained loops.

3.10

Define reference voltage, two node voltages, and one dependent voltage, as shown:



Node A: $\frac{V_A - 0V}{2\Omega} + \frac{V_A - 10V}{1\Omega} + 2A = 0$

Node B: $-2A + \frac{V_B - 10V}{2\Omega} + \frac{V_B - 0V}{5\Omega} = 0$

$$I = \frac{V_B}{5\Omega}$$

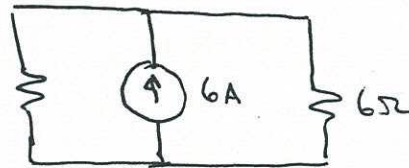
Note: The above solution depends upon my choice of reference node, independent nodes, and dependent voltages. Alternate, valid solutions are possible!

3.11

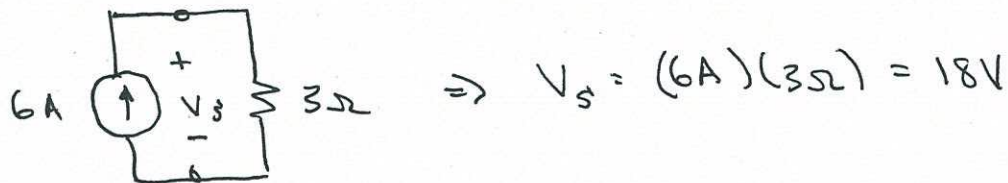
I chose to do the problem using circuit reduction techniques.

The resistances to the left of the 6A source combine to a single equivalent 6Ω resistor:

$$\frac{6\Omega \cdot 12\Omega}{6\Omega + 12\Omega} + 2\Omega = 6\Omega$$

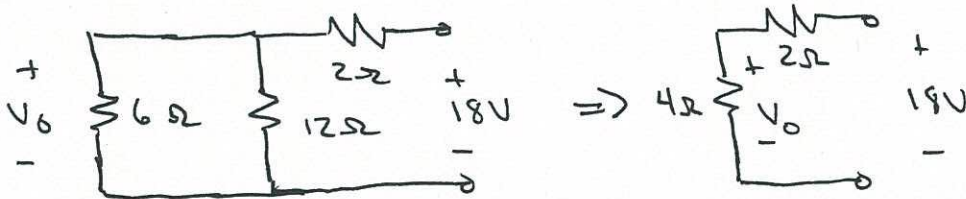


The two parallel 6Ω resistors combine to a single 3Ω equivalent resistance:



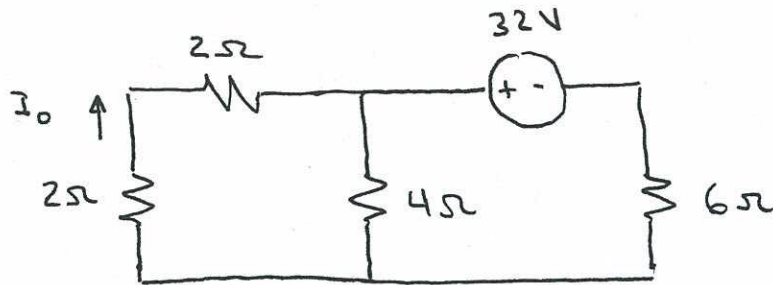
The voltage across the current source is 18V.

we can find V_o from a

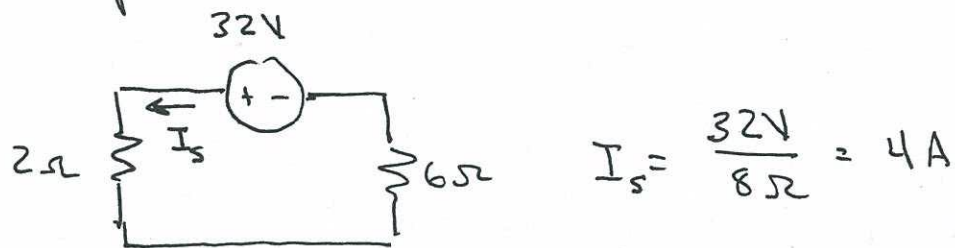


$$V_o = \left(\frac{4\Omega}{4\Omega + 2\Omega} \right) 18V \Rightarrow \boxed{V_o = 12V}$$

3.12



I used circuit reduction. The above circuit is equivalent to:



We can find I_0 from I_s and a current divider formula:



$$I_1 = \left(\frac{4\Omega}{4\Omega + 4\Omega} \right) 4A$$

$$I_1 = 2A$$

$$I_0 = -I_1 \Rightarrow \boxed{I_0 = -2A}$$